

Respiration in Plants

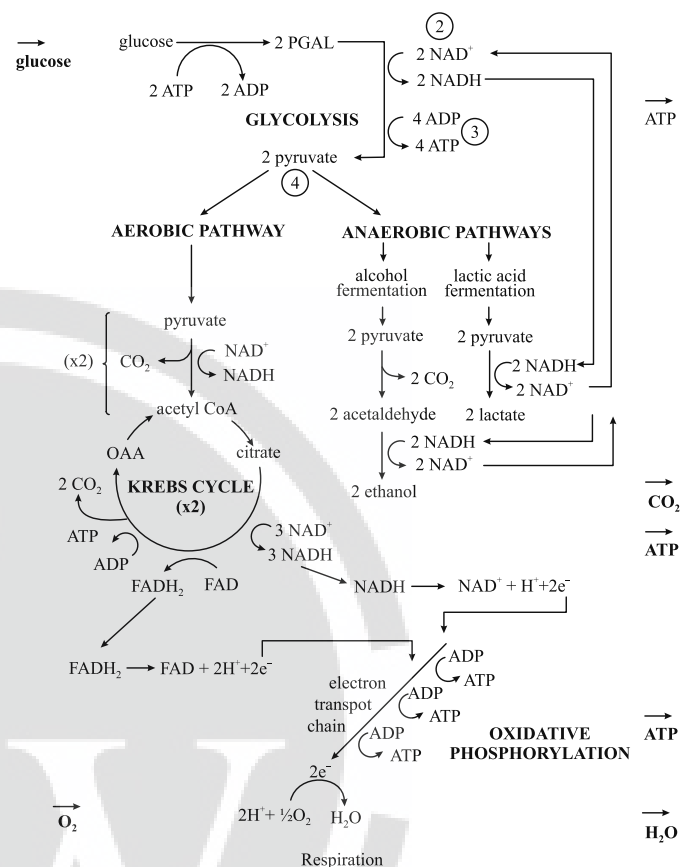
SHORT NOTES

- ❖ Plants do not have special systems for breathing, rather their exchange apparatus: stomata and lenticels are distributed in plant body.
- ❖ Stomata are largely placed on leaves and lenticels on stem.
- ❖ Cellular respiration breaks C-C bonds in glucose molecules to release energy.
- ❖ This energy is either directly used to create ATP (substrate level phosphorylation) or high energy electron carriers- NADH and FADH_2 . These electron carriers further release energy in subsequent steps by oxidative phosphorylation.
- ❖ Cellular respiration can take place with oxygen yielding high amounts of energy (36-38 ATP) or without oxygen yielding only 2 ATP.
- ❖ Fermentation (anaerobic respiration) may be lactic acid fermentation producing lactic acid or alcoholic fermentation producing ethanol and carbon dioxide.
- ❖ Steps of cellular respiration include
 - + Glycolysis: Breakdown of glucose to pyruvate in the cytoplasm and thus producing ATP and NADH.
 - + Conversion of pyruvate to acetyl CoA in mitochondria producing NADH.
 - + Kreb's Cycle: Acetyl CoA enters krebs cycle in mitochondrial matrix, thus releasing carbon dioxide and produces NADH and FADH_2 .
 - + Electron transport chain uses NADH and FADH_2 to produce proton gradient.
 - + Oxidative phosphorylation uses protein gradient and ATP synthase to produce ATP.
- ❖ The table below accounts the ATP production in all steps

Table 5

Source	FADH_2 Produced	NADH Produced	ATP Yield
Glycolysis			2 ATP
Glycolysis		2 NADH	6 ATP
Pyruvate to acetyl CoA		2 NADH	6 ATP
Krebs cycle			2 ATP
Krebs cycle		6 NADH	18 ATP
Krebs cycle	2 FADH_2		4 ATP
Total			38 ATP

A balance sheet accounting for ATP production from glucose by aerobic respiration. Total ATP production can vary between 36 to 38 ATP for each glucose processed.



- ❖ Cellular respiration is an amphibolic pathway. It breakdown glucose. Besides it also is capable of using other substrates like proteins and fats to produce energy. Many of the of the intermediates can be precursors for different synthesis pathways. Like Acetyl CoA can be used to make fatty acids as well.
- ❖ A measure of how good a substrate a substance is, we calculate respiratory quotient, as volume of CO_2 evolved/volume of O_2 consumed. Fats have an RQ < 1, carbohydrates = 1, proteins = 0.9